MODIS Terra and Aqua

Understanding the data for FIREBRIDGE

17 January 2024 to 17 January 2025

The analysis covers 367 UTC dates

Worldwide points, with a northern Morocco study window

Scope	Evidence
Requested dates	17 Jan 2024 through 17 Jan 2025, inclusive
Standard archive	13 monthly MCD14ML files, Collection 6.1
Analysis detections	4,876,343
Dates with global detections	367 of 367
Independent January audit	202,406 shared-field matches

January 2024 contains 15 included dates. January 2025 contains 17.

One row is one thermal detection

A real Aqua example inside the northern Morocco study rectangle

Field group	Example	Meaning
Acquisition	2024-02-21 14:04 UTC	The satellite observation time
Platform	Aqua	MODIS on the Aqua satellite
Pixel center	35.5003, -5.5547	Latitude and longitude in degrees
Brightness	307.6 K / 297.1 K	Thermal band temperatures
FRP	5.5 MW	Radiative power inside the pixel
Quality and class	58 confidence, D, type 0	Native algorithm fields

One ongoing fire can produce many rows. A row does not define a fire perimeter.

Terra and Aqua sample different times

Platform differences include fire evolution and observation conditions

Metric	Terra	Aqua
All detections	1,528,867	3,347,476
Daytime fraction	59.8%	92.8%
Type 0 and confidence ≥30	1,323,943	3,109,700
Median FRP	16.3 MW	20.6 MW
Daytime median solar time	9.91 h	14.20 h

Keep Terra and Aqua separate. FIREBRIDGE uses Aqua as its reference convention.

Data validity and observation coverage

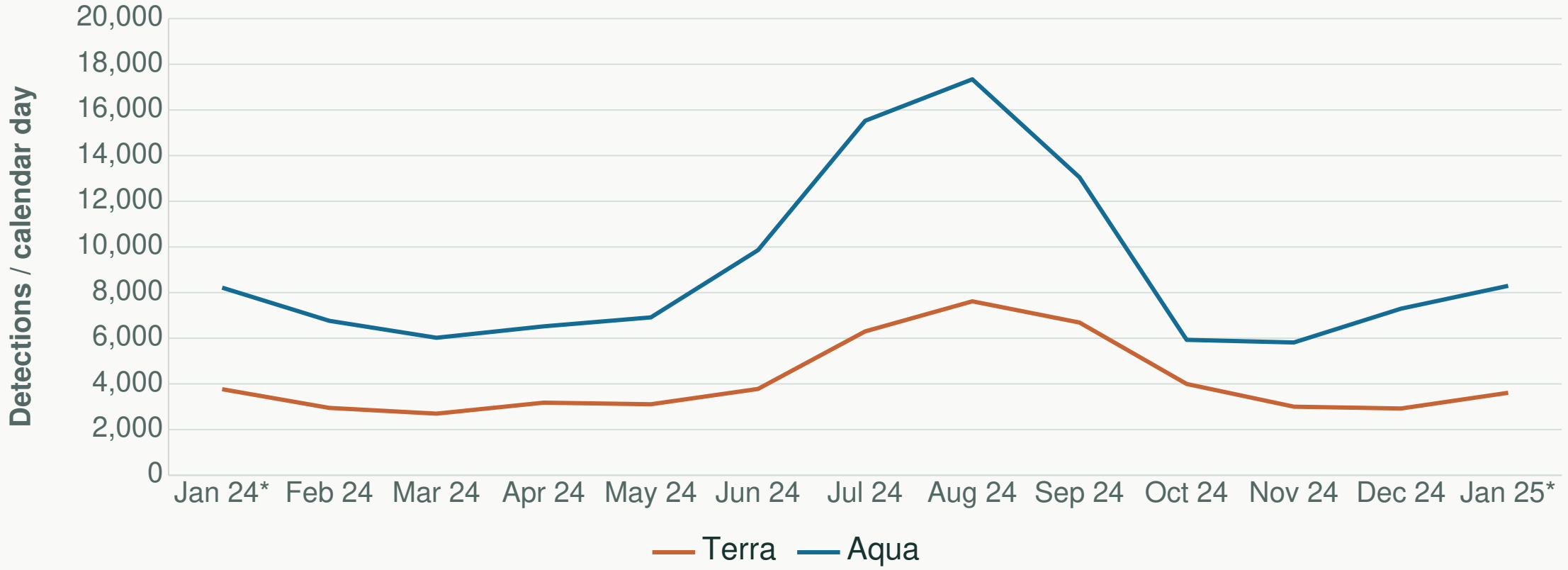
The table checks answer whether fields and records are usable

Check	Rows
Exact duplicate excess	0
Rows quarantined by validation	0
Invalid UTC acquisition	0
Invalid coordinates	0
Negative FRP values	0
Zero FRP values	2,425

January audit: 100.0% of UMD rows match all shared fields.

Global recorded activity peaks in Aug 24

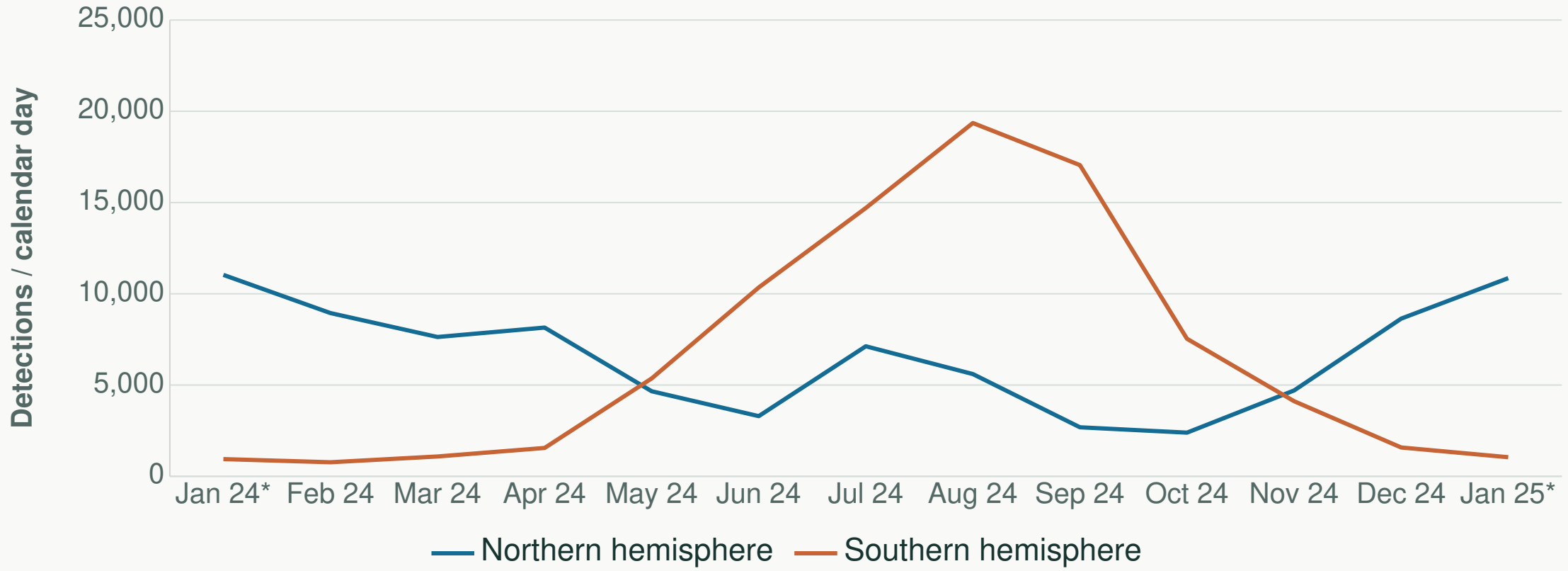
All detections per included calendar day, by acquisition month



* Partial January windows. Rates describe recorded detections without a coverage correction.

Northern and southern patterns differ

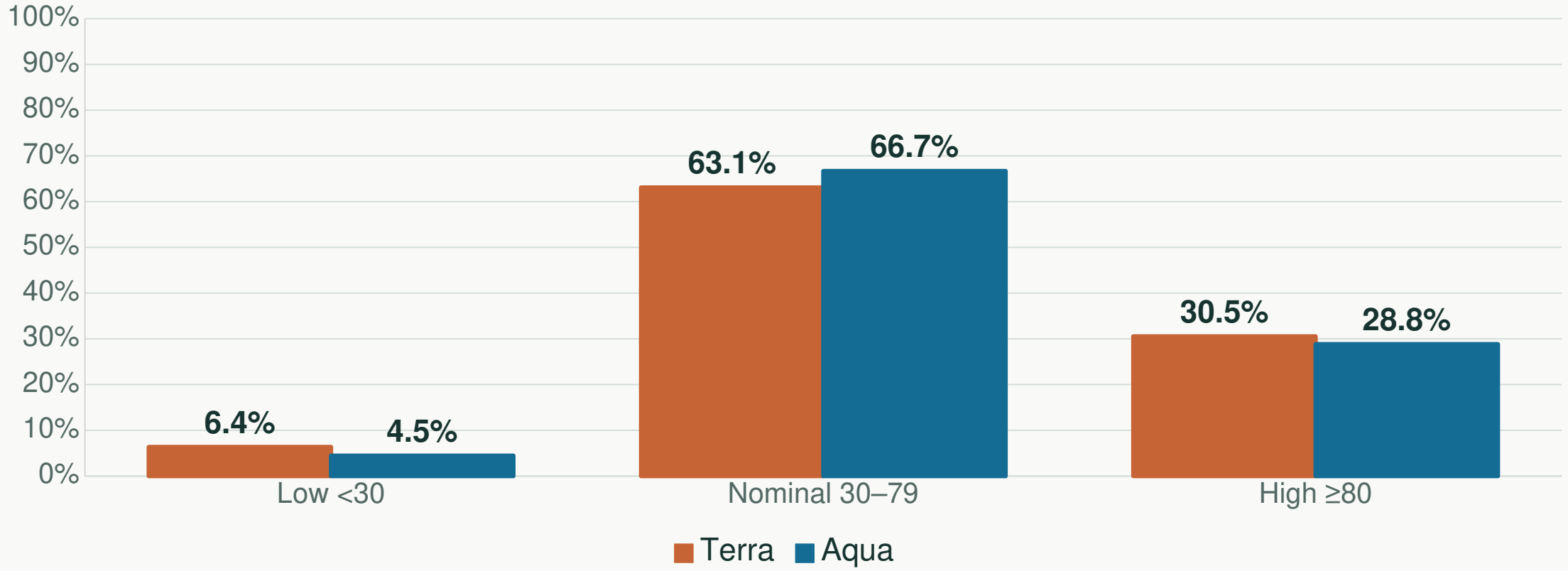
Both platforms combined, all detections per included calendar day



Global totals combine regional calendars. The report includes spatial bins for closer inspection.

Confidence changes which rows we retain

Native score classes, shown as a share of each platform



Project baseline: type 0 with confidence ≥30. Keep all and ≥80 views for sensitivity.

FRP has a long upper tail

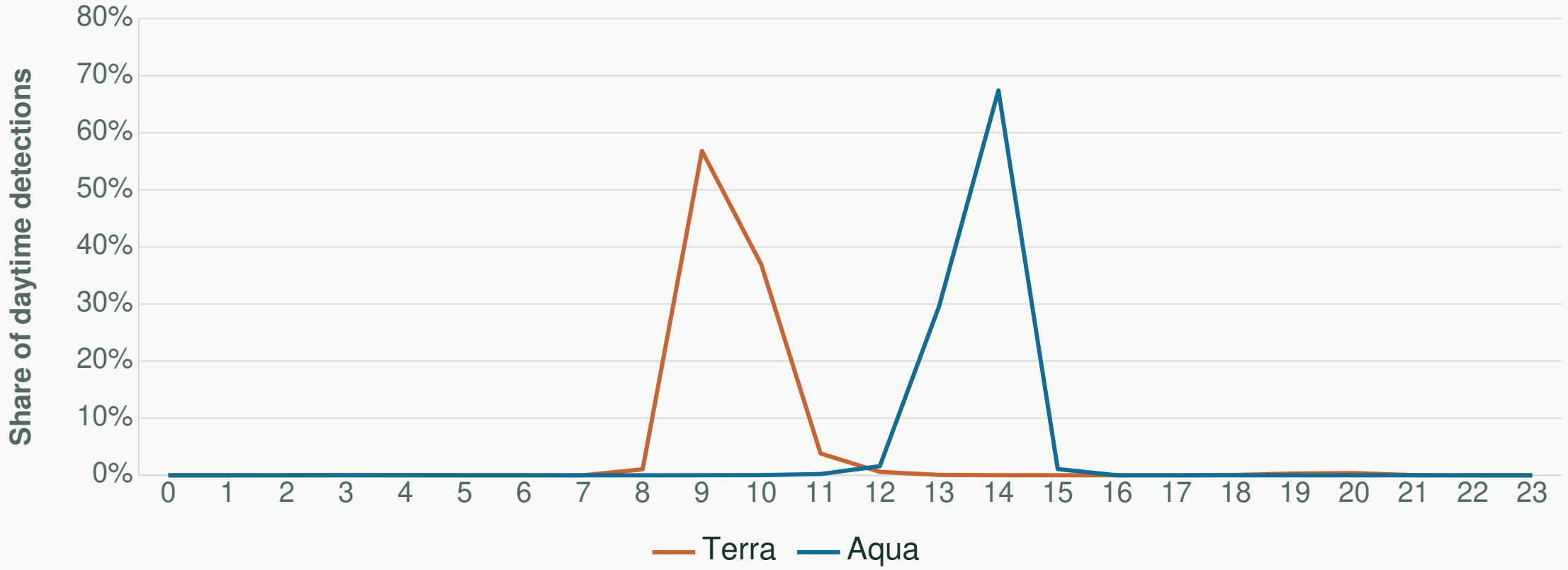
Exact quantiles across each platform, in megawatts

FRP summary	Terra	Aqua
Median	16.3	20.6
95th percentile	115.7	143
99th percentile	337.5	408.7
Maximum	12,423.7	13,490.3
Zero FRP rows	1,472	953

FRP measures power at acquisition time. Negative values are excluded from these quantiles.

Daytime detection times show separate regimes

Approximate local solar hour, conditional on a recorded daytime detection



Solar hour ≈ UTC hour + longitude / 15. It differs from civil time and orbit crossing time.

Pixel footprints vary across the swath

Measured FIRMS dimensions for 1–17 January 2025 only

Dimension	Terra	Aqua
Median scan width	1.2 km	1.2 km
95th percentile scan width	3.3 km	3.1 km
Median track height	1.1 km	1.1 km
95th percentile track height	1.7 km	1.7 km
Rows in this overlap	61,396	141,010

The full MCD14ML window has a sample index, with no explicit scan/track dimensions.

Thermal anomalies have inferred source types

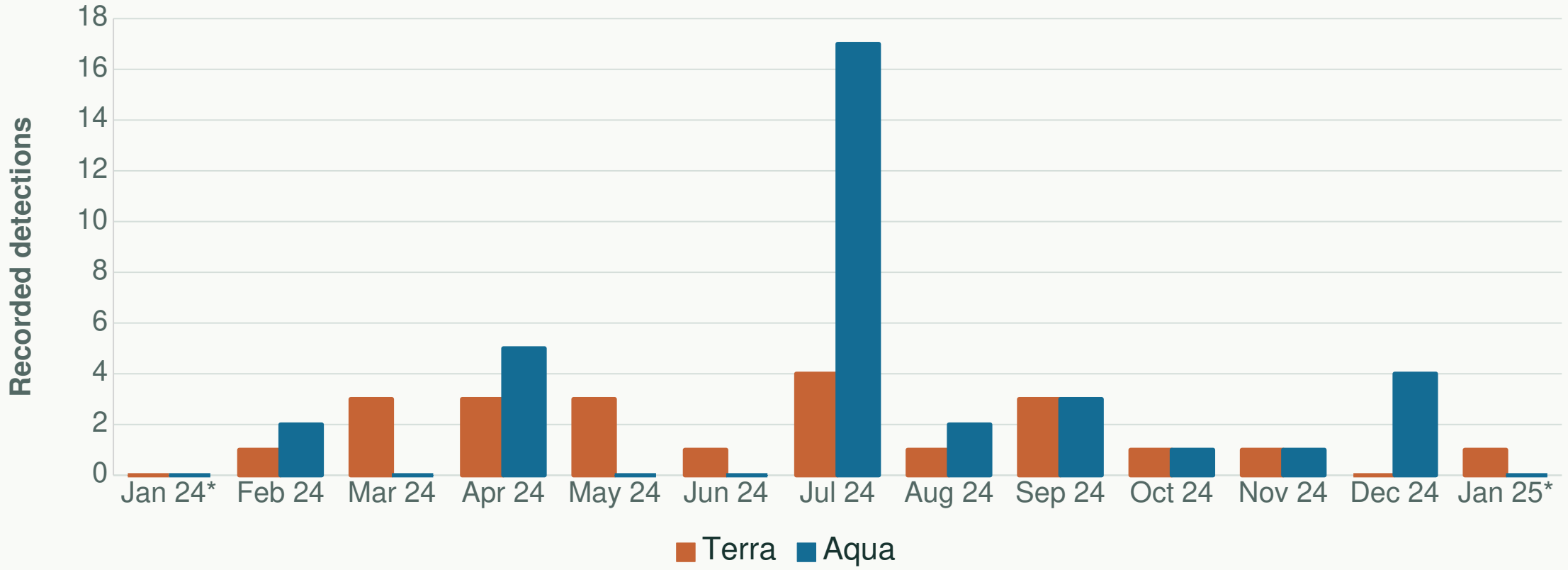
Native type codes support review, with no incident-level verification

Type	Terra detections	Aqua detections
0: Presumed vegetation	1,411,637	3,252,331
1: Active volcano	4,221	3,609
2: Other static land	103,186	82,900
3: Offshore	9,823	8,636

Type 0 can include agricultural burning. Recurrence alone does not justify deletion.

The Morocco pilot contains 57 detections

49 meet type 0 and confidence ≥30 inside the study rectangle



Bounds: 6.4°W to 4.5°W, 34.5°N to 35.9°N. This is a study rectangle, not all Morocco.

The FIREBRIDGE response

A supported Aqua-equivalent detection indicator in a 5 km cell

Reference: at least one retained Aqua detection in a supported cell and pass.

VIIRS provides the prediction features for a withheld-Aqua experiment.

Terra keeps a separate adapter because its observation regime differs.

The EDA prepares the native record. Observation support defines the labels.

Observation masks make absence labels possible

A point table only describes detected thermal anomalies

Cell/pass state	Evidence needed
Retained detection	Point detection plus the defined inclusion rule
Usable observation with no retained detection	Full fire mask, QA and geolocation
Cloud, invalid or missing observation	Full mask and an explicit exclusion reason
Aqua science files	MYD14 plus collection-compatible MYD03
Terra science files	MOD14 plus collection-compatible MOD03

An archive with rows on every date does not prove that every cell had a usable observation.

The next deliverable is a supported pilot table

A small reproducible season and area before model training

1. Retrieve a manageable season of matched fire masks and geolocation.

2. Build cell/pass support with usable-land fractions and exclusion reasons.

3. Evaluate the proposed 5 km, 30-minute and 80% shared-land rules.

4. Freeze a transparent baseline, then test with withheld time and geography.

Deliver the native observations, support flags and baseline errors together.